Community Project Proposal
SE6101

University Bursary

Management System

Group-07

Department of Software Engineering

Faculty of Computing

Sabaragamuwa University of Sri Lanka

February 2026

University Bursary Management System

Approval of Community Project

1. Group Details:

Index Number	Name with Initials	Email	Mobile Number
21CSE0157	W.P.I.N.Kumarasinghe	wpinkumarasinghe@std.foc.sab.ac.lk	0765735091
21CSE0158	L.B.Madhuranga	lbmadhuranga@std.foc.sab.ac.lk	0763546936
21CSE0171	Ulindu Dakshitha	wmudbandara@std.foc.sab.ac.lk	0717028242
21CSE0184	W.M.R.R.Wijekoon	wmrrwijekoon@std.foc.sab.ac.lk	0761098381
21CSE0224	K.A.K Dinal	kakdinal@std.foc.sab.ac.lk	0702236865

2. Title of the Project : University Bursary Management System
3. Name of the Supervisor : Ms. S Sweshthika
4. Supervisor's designation : Lecturer (Prob.)
5. Supervisor's Organization : Faculty of Computing,
   Sabaragamuwa University of Sri Lanka

For office use only:

6. Approved/Not approved :
7. Date :
8. Suggestions if any :
9. Signature of Internal Supervisor :

ACKNOWLEDGEMENT

To begin with, we wish to state that we are extremely grateful to the Community Project Coordinator and Dean of the Faculty, Prof. S. Vasanthapriyan, to have been so supportive and helpful to the project in terms of leadership, persistent encouragement and guidance. His leadership and dedication to education excellence have been a big source of inspiration to us in achieving success in implementing this initiative.

We wish to thank the Head of the Department of Software Engineering, Prof. B. T. G. S. Kumara, with all our hearts, for his guidance and support. His guidance, coupled with his assistance in introducing us to the Student Service Center and the Senior Assistant Registrar gave us a precious chance of conducting an efficient requirement analysis of this project.

We are also very thankful to Senior Assistant Registrar, as well as staff of Student Affairs Division, who provided us with the chance to study and learn the current process of managing bursary. They played a vital role in the system requirements and in determining the feasibility of our solution by their cooperation and input.

This is due to the assistance of Mrs. Samadhi who is an active bursary manager in the Faculty of Computing and who has been of constant assistance as well as providing domain expertise and being open to offer real life challenges and processes. Her work provided a great insight into the system.

We also wish to offer our thanks to our supervisor, Ms. Sweshthika, because of the consistent guidance, helpful feedback, and encouragement that she provided during the process of this project development. Her mentoring was also important in polishing our ideas and ensuring our work is of high quality.

Lastly, we would like to give our thanks to all the people who either assisted us directly or indirectly in this project. They have provided the necessary support, collaboration, and input to ensure the completion of this work is successful.

Thank You,
Group 07

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I. GROUP DETAILS AND CONTRIBUTION

Index Number	Name with Initials	Contribution
21CSE0157	W.P.I.N.Kumarasinghe	UI/UX, Frontend Development , Backend Development
21CSE0158	L.B Madhuranga	UI/UX, Frontend Development , Backend Development
21CSE0171	Ulindu Dakshitha	UI/UX, Frontend Development , Backend Development
21CSE0184	W.M.R.R.Wijekoon	UI/UX, Frontend Development , Backend Development
21CSE0224	K.A.K.Dinal	UI/UX, Frontend Development , Backend Development

1) INTRODUCTION & OBJECTIVES

1.1 Introduction

In the digital age, effective and open management of financial assistance systems is very important to educational institutions. The bursary management procedure within the universities usually comprises several departments, paper work, and complicated approval procedures that may cause delays, miscommunication, and invisibility to the students.

In order to overcome such obstacles, we suggest that a University Bursary Management System be developed, which is an online system that will be centralized to make the process of handling bursarial tasks simpler and fully automated. Such a system will make the coordination between students, faculty, and officers at the bursary department to be smooth and correct in terms of processing bursary allowances in time.

The system will allow students to verify the eligibility, payment history, and to raise concerns through the platform. Concurrently, faculty and administrative staff will be in a position to check attendance, approvals and process data on bursary quickly and conveniently owing to role-based access.

One of the main characteristics of this system is the ability to track the workflow of the attendance sheets in real-time so that all stakeholders are able to view the current state and location of documents in the approval chain. The system will also contain an academic calendar module, which will allow faculties to publish their schedules to make attendance verification and transparency possible.

The system will be user friendly, have strict access control and responsive design that will allow the system to be offered to various devices to make it convenient to all users. This is a digital solution that seeks to improve the efficiency of the operations, manual workload, and transparency in managing the bursary within the university.

1.2 Objectives

General Objectives

- Create user-friendly and centralized system in order to organize bursary processes effectively.
- Enhance trust and interaction between students, faculty and bursary.
- Minimize manual errors and delays of the traditional paper-based systems.

Specific Objectives

- Create a user friendly and intuitive web based system that will provide easy navigation and comfort to every stakeholder.

- Install a role-based workflow system that handles the bursary approval system at various levels such as faculty and administrative divisions.
- Offer a tracking system that users can use in real-time to keep track of the summary and position of the attendance sheets in the approval process.
- Provide the students with an integrated communication system that allows navigating through bursary eligibility, payment history and submitting issues or queries.
- Develop and deploy an academic calendar module where faculties would be able to post schedules and share schedules that would assist in the process of attending to schedules.
- Make sure the system is secure by means of authentication and access control, sensitive student and financial data is secured.
- Installing a comprehensive responsive platform that will perform well on desktops, tablets, and mobile devices.

2) ANALYSIS

2.1 Feasibility Study

2.1.1 Technical Feasibility

University Bursary Management System Technical feasibility: It entails assessment of the tools, technologies, and infrastructure to be used in successful development, deployment, and maintenance of the University Bursary Management System. The system is so made that it offers to provide secure data exchange, effective workflow, and free flow of interaction among various roles of the users such as the students, the members of the faculty, and the bursary officers.

The suggested system embraces a scalable and modern technology stack, which complies with the existing server environment of the university and allows upcoming improvement.

Technology Stack:

- ❖ Front-End:

Next.js with TypeScript to create a responsive and user-friendly fast interface. This allows making the application efficient, performance-wise, and more maintainable.

- ❖ Back-End:

Express.js provides a lightweight, scalable, and secure framework for server-side development. It offers flexible middleware architecture, efficient API handling, and smooth management of complex processes such as multi-level approval workflows and authentication.

- ❖ Database:

MySQL Server to have a good data storage. It guarantees effective management of relationship data like student records, attendance sheets, approvals and transactions.

- ❖ API Communication:

RESTful APIs will be adopted to facilitate the effective interaction between the back and frontend systems.

System Capabilities

- Effective management of multi-role processes (Student, Faculty MA, HOD, AR, SAR, Bursary MA).
- Real time follow-up of attendance sheets status in departments.
- Role-based access control and secure authentication.
- Bursary-related information is centrally managed.

Scalability:
System architecture will support future development and improvement which includes:

- More users (increasing number) (students and staff).
- New modules like payment integration or reporting will be added.
- Connection to other university systems (e.g. student information systems).
- Growth to accommodate mobile apps or other platforms.

Express.js and Next.js mean that the system will be maintainable, scalable, and suitable to the changing needs of the institution.

2.1.2 Economic Feasibility

Economic feasibility judges the cost-effectiveness of the proposed bursary management system to be developed and maintained.

- The technologies implemented such as Larval, Next.js, and MySQL are open-source and can be used freely which tremendously minimizes the cost of software licensing.
- The system will be installed with the help of the available university infrastructure which will reduce hosting or deployment costs.
- There is no extra cost incurred by students or the staff to access the system and therefore it is accessible by everyone.

Additionally, the system is cost-effective because it:

- Minimizes administrative waste and paperwork.
- Reduces processing time and faults.
- Enhances effectiveness in the bursary management processes.
- Conserves time and resources to students and administrative staff.

All in all, the suggested system is very cost-effective as well as long-term.

2.1.3 Operational Feasibility

Operational feasibility evaluates whether the **University Bursary Management System** will function effectively within the university environment, ensuring a smooth, efficient, and transparent bursary management process for all stakeholders.

User-Friendly Interface

The system will have a simple and user-friendly interface that will enable users who will be students of the institution, faculty members and bursary officers to use the platform with minimum effort.

- Bursary eligibility, the payment history and the account of the status of the applications can be easily viewed by the students.
- Role-based dashboards allow staff members to effectively manage the verification of attendance, approvals and workflow.
- The responsive design is user-friendly because it is available on different platforms such as desktops, tablets, and smartphones, and all users will find it convenient.

Impact on Operations

The given system is a great enhancement to the current process of managing bursaries adding automation and electronic processing.

- Automates the workflow of checking and approving attendance of various personnel, which includes Faculty MA, HOD, AR, SAR, and Bursary MA.
- Allows tracking the attendance sheets in real-time so that users can determine the current status and position in the approval process.
- Gets a centralized way of communicating with the use of an issue/ticket system, where people do not need to make physical visits and manual inquiries.
- Allows faculties to post academic calendars, which adds to correct attendance validation and enhances the interaction of departments.
- Eliminates the necessity to maintain paper-based records, minimizing the workload of the administration and decreasing human errors.
- Ensures the accuracy and consistency of data through the digital control of approvals, student records and bursary transactions.

2.2 Data Flow Diagram

2.2.1 Context Diagram

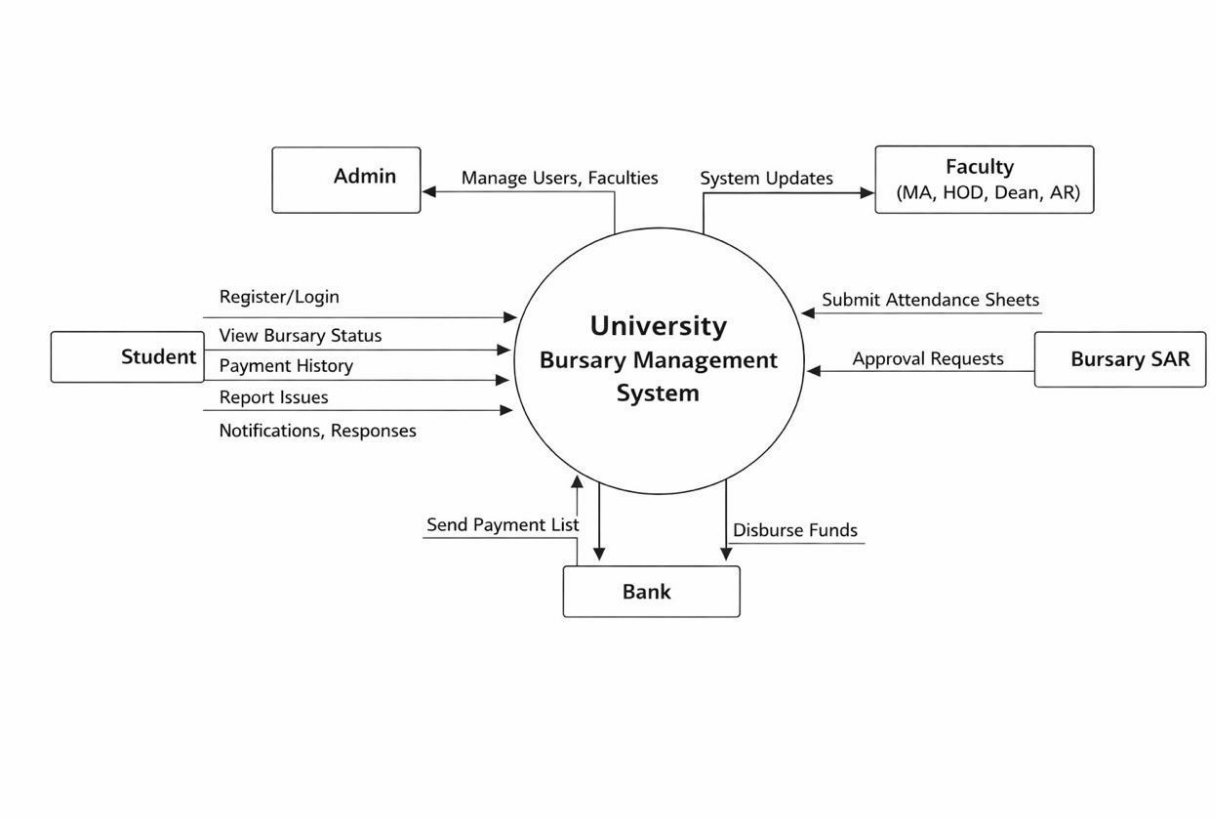


Figure 1 Context diagram

This context diagram indicates the information flow in the University Bursary Management System. Through the system, students are able to apply on bursary, see eligibility and payment information and raise concerns. Faculty and administrative officers deal with the process of attending verification, approvals, and updating of workflow, whereas the bursary department deals with bursary records and bursary processing. The system also assists in communication with the Senior Assistant Registrar as it gives reports, receives updates on policies, and ensures that there is smooth coordination of all those involved in the bursary management system.

2.3 ER Diagram

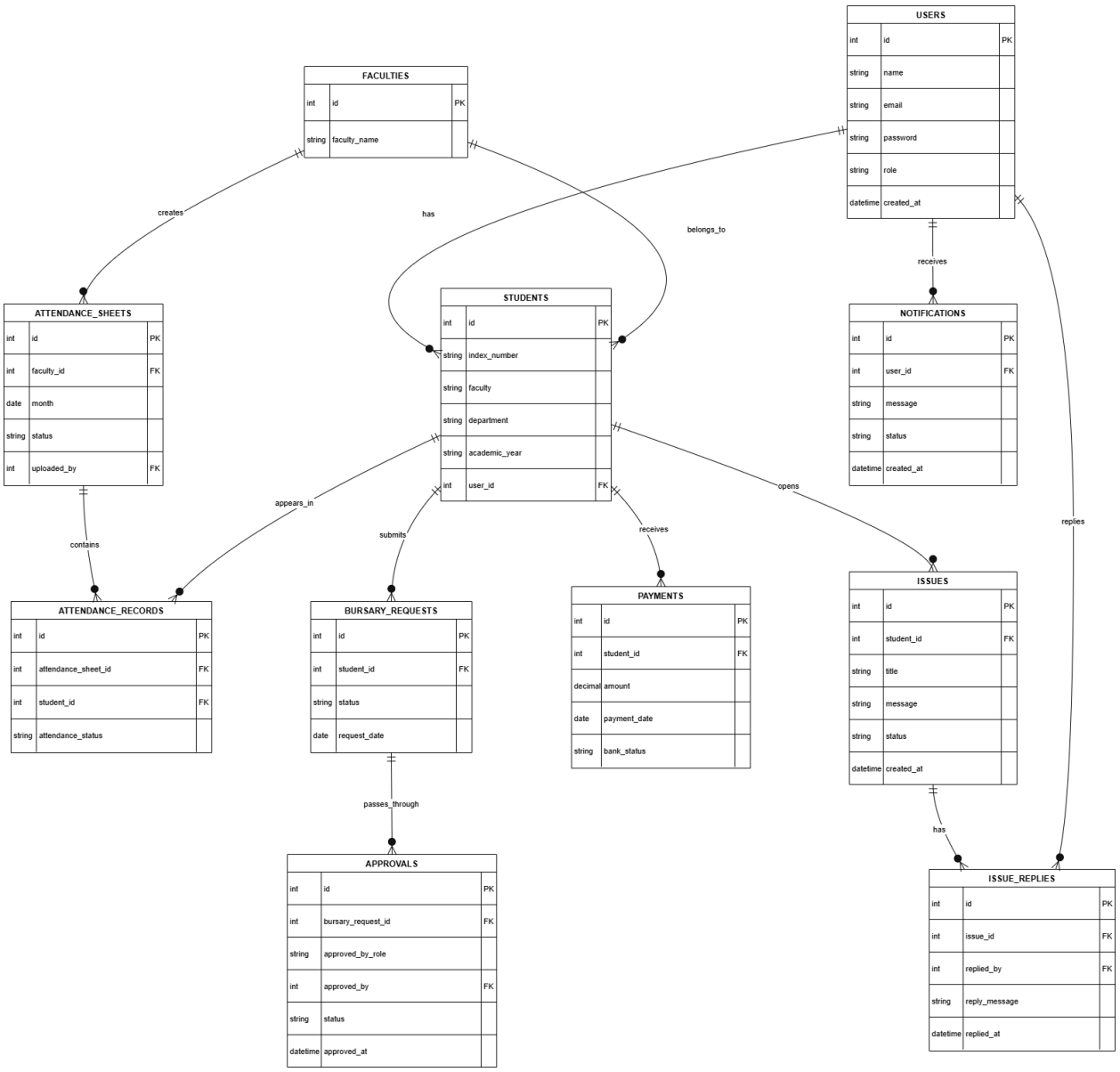
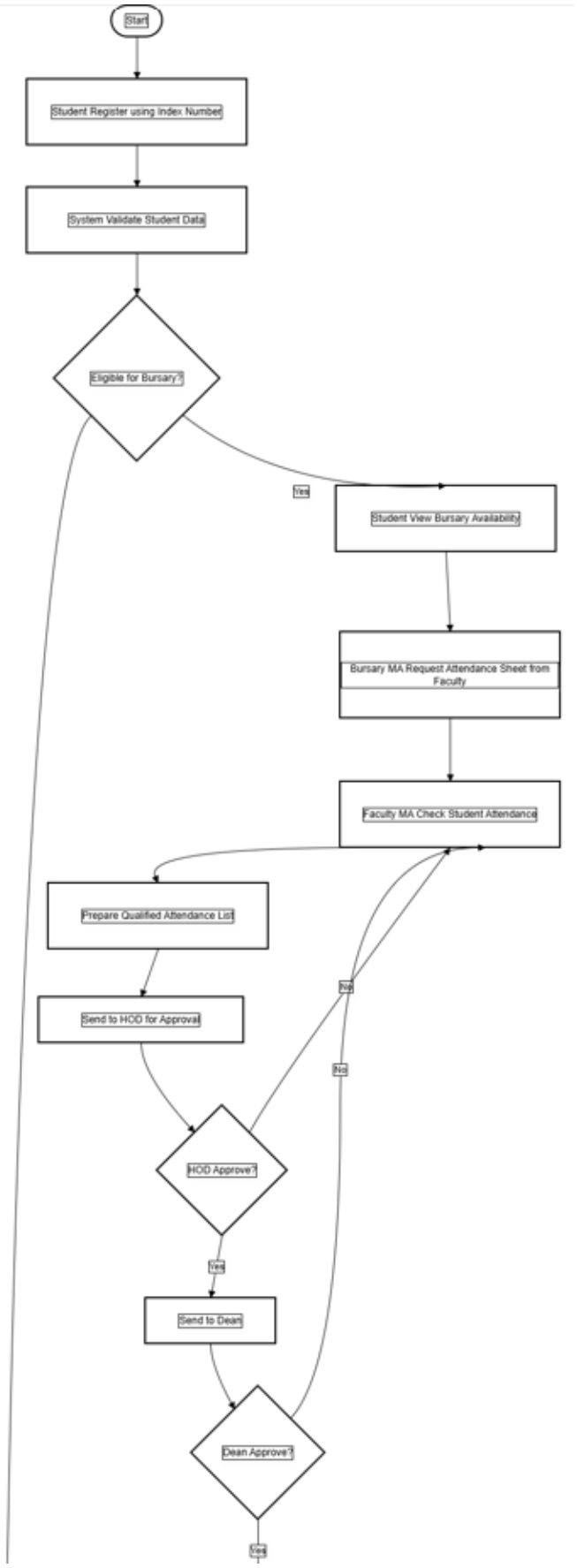


Figure 2 ER diagram

This ER Diagram is that of the database of the University Bursary Management System. Students are able to register, see their bursary eligibility, track their payment history, and raise issues, whereas administrative users Bursary MA, Student Service SAR, Faculty AR, Department MA and HOD are able to look at attendance checking, approval processes, and bursary process. The main objects are Student, User, Bursary Application, Attendance Sheet, Approval Workflow, Issue Ticket, Payment Record, and Academic Calendar with the attributes that include student details, approval status, workflow state, payment information and uploaded academic schedule. The diagram graphically represents the data storage, management and interrelation in the system to facilitate an effective transparent process of managing the bursary.

2.4 Flow Chart



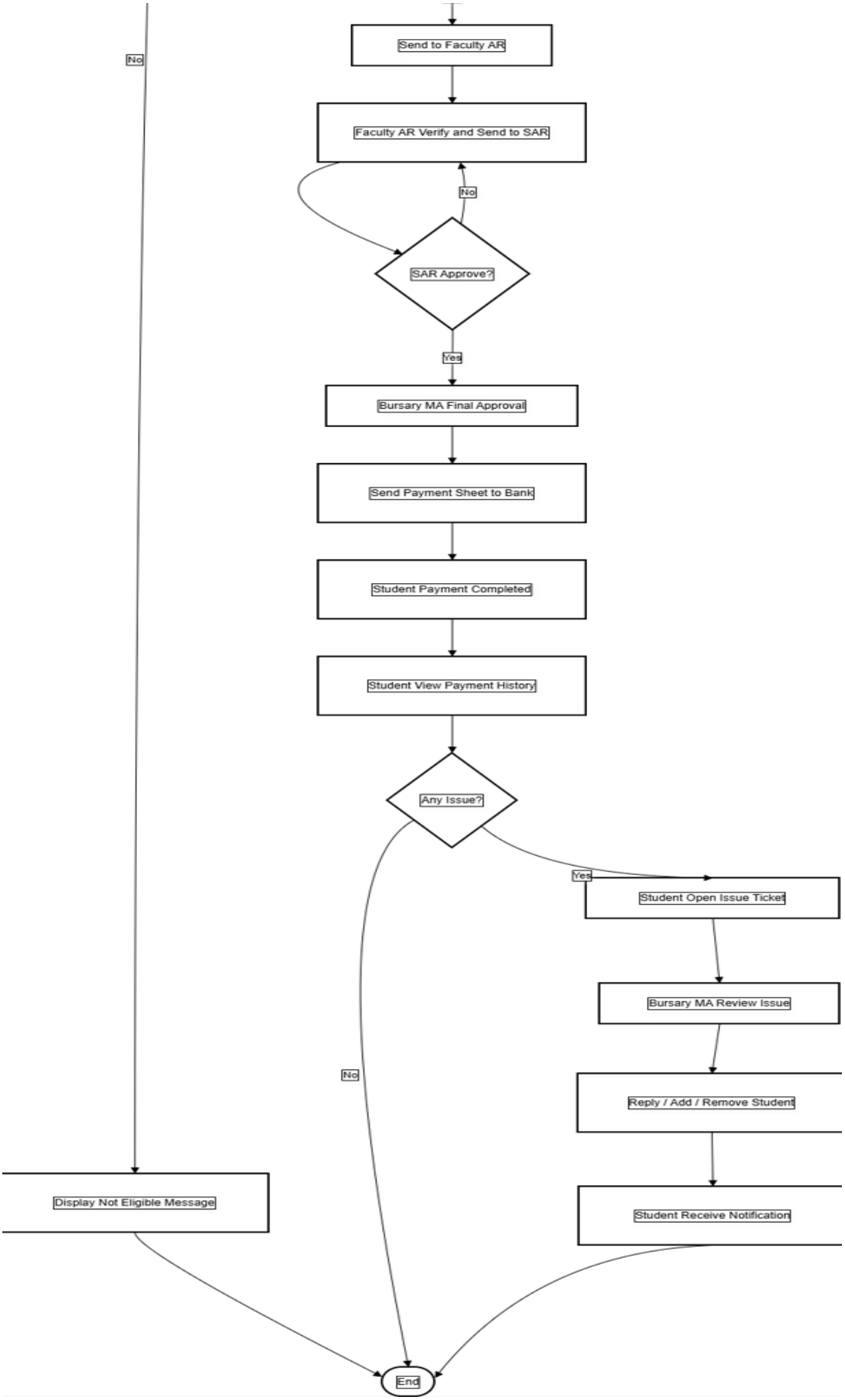


Figure 3 Flow Chart

This flow chart shows the process of bursary management at the University Bursary Management System. Students enter the system with their index number and make sure that they are eligible. When qualified, their information is sent on the attendance verification process, beginning with the Bursary MA to Student Service SAR, Faculty AR, Department MA and HOD to approve the information. The attendance sheet is examined and sent by each party. Upon approval, the process goes in reverse through Faculty AR and Student Service SAR to Bursary MA to be processed finally. The status at every stage can be monitored by the students and the issue may be raised. Workflow is terminated in case of non-meeting of any eligibility or when the process is rejected at any point.

2.5 Use Case Diagram

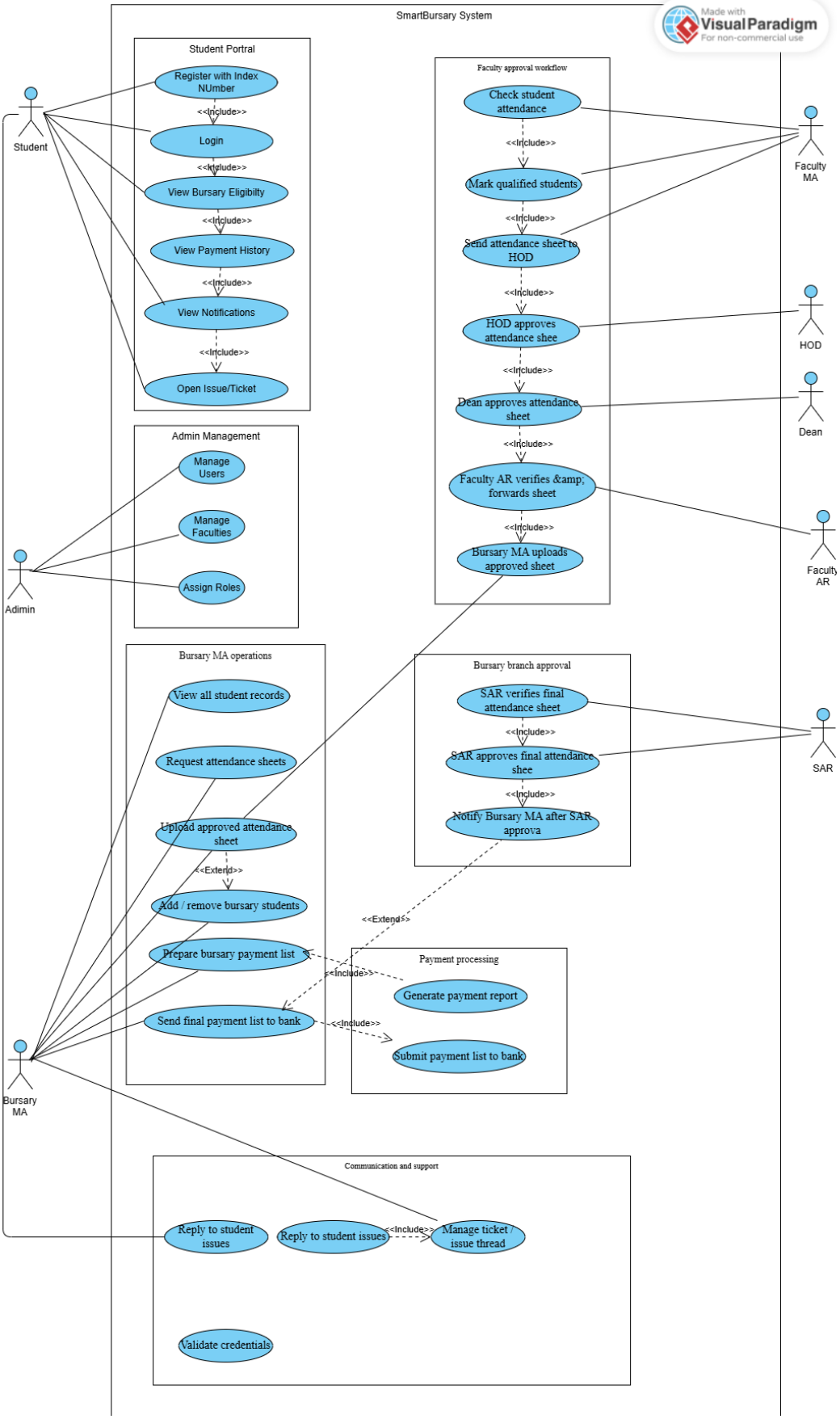


Figure 4 Use Case Diagram

The diagram below is a Use Case Diagram that represents the features of the University Bursary Management System. Students are able to register, log in, check the eligibility of the bursary, check the history of payment, track the status of the attendance sheet, and raise issues or ask questions. The established workflow allows faculty members and the department officers to verify attendance, update records, and grant requests. The workers in the bursary department handle student records, give approvals and address complaints. The Senior Assistant Registrar is in charge of approvals and policies. The flowchart presents a pictorial representation of how the system interacts with the users..

3) HARDWARE AND SOFTWARE REQUIREMENTS

3.1 Hardware Requirements

- **Processor:** Intel Core i5 (8th Generation or higher) / AMD Ryzen 5 equivalent
- **RAM:** Minimum 8GB RAM (16GB recommended for smooth development and multitasking)
- **Operating System:** 64-bit Operating System (Windows 10/11, Linux, or macOS)
- **Storage:** Minimum 256GB SSD (512GB SSD recommended for better performance)
- **Internet Connection:** Stable broadband internet for package installation, updates, Git operations, and deployment
- **Display:** Minimum Full HD monitor (1920×1080 recommended)
- **Peripherals:** Keyboard and mouse
- **Additional Devices:** Smartphone / Tablet for responsive interface testing

3.2 Software Requirements

3.2.1. Main Software Requirements

- WebStrome - Main IDE for backend development
- Visual Studio Code - Lightweight code editing and frontend support
- draw.io - ER diagrams, flowcharts, architecture diagrams
- Google Docs - Documentation writing
- Figma - UI/UX design
- Git - Source control
- GitHub - Repository hosting
- Jira - Project Management

3.2.2. Technologies

- Express.js
- MySQL with Sequelize orm
- Next Js
- Tailwind Css

4) TABLE AND STRUCTURE, NUMBER OF MODULES, DETAILS OF MODULES, DATA STRUCTURE

4.1 Table and Structure

Student

- student_id (Primary Key)
- index_number
- first_name
- last_name
- email
- phone_number
- faculty
- department

Bursary_MA

- bursary_ma_id (Primary Key)
- name
- email
- password

Student_Service_SAR

- sar_id (Primary Key)
- name
- email
- password

Faculty_AR

- faculty_ar_id (Primary Key)
- name
- email
- faculty

Department_MA

- department_ma_id (Primary Key)
- name
- department
- email

HOD

- hod_id (Primary Key)
- name
- department
- email

Bursary_Record

- bursary_id (Primary Key)
- student_id (Foreign Key → Student)
- eligibility_status
- payment_status
- amount
- issued_date

Attendance_Sheet

- sheet_id (Primary Key)
- student_id (Foreign Key → Student)
- faculty
- attendance_percentage
- status
- current_stage
- last_updated_date

Issue_Ticket

- ticket_id (Primary Key)
- student_id (Foreign Key → Student)
- subject
- message
- reply
- status
- created_date

Academic_Calendar

- calendar_id (Primary Key)
- faculty
- event_name
- start_date
- end_date
- description

4.2 Number of Modules

1. User Management
2. Bursary Eligibility & Status Management
3. Attendance Workflow Management
4. Issue / Ticket Management
5. Academic Calendar Management

4.3 Details of Modules

User Management

- **Functions:** Student registration, login, role-based access control
- **Data Structures:** User details stored in Student and Staff tables
- **Components:** Login forms, authentication system, role-based dashboards

Bursary Eligibility & Status Management

- **Functions:** View eligibility, track payments, update bursary records
- **Data Structures:** Bursary data stored in the Bursary_Record table
- **Components:** Student dashboard, status view interface

Attendance Workflow Management

- **Functions:** Manage attendance sheet approval workflow across multiple roles
- **Data Structures:** Attendance data stored in the Attendance_Sheet table
- **Components:** Workflow tracking system, approval interfaces

Issue / Ticket Management

- **Functions:** Students can raise issues, and bursary staff can respond
- **Data Structures:** Ticket data stored in the Issue Ticket table
- **Components:** Issue submission form, communication panel

Academic Calendar Management

II. **Functions:** Faculties upload and manage academic calendars
III. **Data Structures:** Calendar data stored in the Academic Calendar table
IV. **Components:** Calendar display interface, update panel

1.1 Data Structure

User Data

Has student and staff account information, authentication and role-based access.

Bursary Data

Stores student eligibility, payment and bursary status.

Attendance Data

Keeps attendance records and follows the stages of workflow approval.

Issue Data

Stores contact between students and the bursary department in terms of tickets.

Academic Calendar Data

Holds faculty and schedules that are used to verify attendance and plan.

2) PROPOSED SYSTEM

2.1 Functional Requirements

Student Functional Requirements

2.1.1 Student Registration & Authentication

- Students are allowed to enroll to the system with their university index number and the personal information they have to provide.
- Students are able to log in with their credentials with a lot of security.
- Students are able to change or reset their password where need be.
- The students are able to change their details in the profile including contact details and email address.

2.1.2 Bursary Eligibility & Status Viewing

- The system allows students to check their status regarding the bursary entitlement.
- Students are able to access the history of bursary payment and other associated information.
- Students will be able to see the notifications or updates of their bursary process.

2.1.3 Attendance Sheet Tracking

- Students will be able to monitor the position of the attendance sheet in which they are filing bursary.
- The students are able to access the present level or point of the attendance sheet in the approval workflow.
- When the attendance sheet is rejected, approved or returned with corrections, students are able to get updates.

2.1.4 Issue / Inquiry Management

- Through the system, students can increase issues, complaints, or queries in regard to matters of bursaries.
- Students will be able to see the replies of the bursary department.
- The students are able to check the status of issued tickets or issues.

2.1.5 Academic Calendar Viewing

- Academic calendars of the faculty can be viewed by students through the system.
- The information of the calendar helps the students to comprehend the academic schedules of the attendance verification.

Administrative and Staff Functional Requirements

2.1.6 Staff Authentication & Access Control

- The employees are required to have secure role-based access control to log in.
- Various tasks like Bursary MA, Student Service SAR, Faculty AR, Department MA and HOD should have only the authorized system functions.
- The system is supposed to be able to provide safe session management and regulated access to confidential bursary information.

2.1.7 Student Data Management

- The student records that are pertinent to bursary processing can be accessed by authorized personnel.
- Bursary department employees are allowed to add, alter or maintain the student related bursary information where it is necessary.
- Employees have the capability of checking and ensuring the accuracy of student data to be processed.

2.1.8 Attendance Workflow Management

- MA Bursary has the ability to start and issue attendance sheets into the working process.
- Faculty AR, Student Service SAR, Department MA, and HOD can check, counter, denial, and send attendance sheets depending on the role.
- The system is able to capture every stage of the workflow and adjust the present status of the attendance sheet.
- The administrator can have the ability to monitor the entire flow of the attendance sheet during the process.

2.1.9 Issue / Ticket Response Management

- All issues and inquiries posted by the students can be viewed by bursary department staff.
- Staff members are permitted to reply to the tickets and change their status.
- The system is able to keep a communication history among the students and staff.

2.1.10 Academic Calendar Management

- The management of the academic calendar is to be done by the two superintendents of instruction, working in consultation with the school board and the school administrators.
- The system offers faculty and staff the opportunity to upload, update, or maintain academic calendars.
- The student and the concerned administration can have access to published academic calendars.
- The updated information on the calendar can be stored and displayed in a structured form in the system.

2.1.11 Bursary Record & Payment Management

- The staff of the bursary department is able to handle bursary eligibility records and information concerning payment.
- Bursary approval outcomes and payment status update on authorized personnel.

2.2 Non-Functional Requirements

Usability

- For students, professors, staff, and bursary officers, the user interface should be simple to use.
- The technology should lessen user misunderstanding and streamline complicated financial operations.
- For use on PCs, tablets, and smartphones, the design should be responsive and optimized.

Accessibility

- Whenever feasible, the system should be built with accessibility requirements in mind.
- All users should be able to access crucial content, and text should be readable and clear.
- Keyboard-friendly navigation will enhance usability for users with various accessibility requirements.

Performance

- For a seamless user experience, the system should load in a fair amount of time.
- When examining records, checking process status, and retrieving student data, the platform should be geared for speedy response times.
- Multiple users should be supported concurrently by the system without experiencing appreciable performance impact.

Security

- Sensitive user data, such as bursary records and student information, should be protected by the system using safe authorization and authentication procedures.
- To stop unwanted access, role-based access control should be implemented.
- Passwords and private information should be handled and maintained safely.

Accuracy

- The system needs to offer precise and current information about bursary eligibility.
- The current state of the approval procedure should be shown in real time in the attendance
- Student information and payment histories should be kept accurate and consistent.

Reliability

- During operating hours, the system should be accessible with little downtime.
- To minimize user interruption, scheduled maintenance should be managed carefully.
- Without losing crucial data, the platform should consistently save and retrieve records.

Scalability

- Future growth, such as more user roles, more modules, and more student data, should be supported by the system architecture.
- The platform should be able to be updated and modified to meet the needs of the university in the future.

2.3 Methodology

The Agile Development Methodology is employed to construct the University Bursary Management System, thereby facilitating adaptability, iterative development, and the incorporation of continuous feedback throughout the project lifecycle.

The development process is organized into the following fundamental phases:

- Requirement Gathering – Systematically identifying the essential requirements, workflow specifications, and expectations of both students and administrative personnel.
- System Design – Formulating database schemas, user interface/user experience designs, workflow architectures, and delineating the overall system architecture.
- Implementation – Executing the frontend development utilizing Next.js in conjunction with TypeScript, the backend utilizing Express.js, and the database implementation employing MySQL Server.
- Testing – Conducting unit testing, integration testing, and user acceptance testing to ascertain both functionality and dependability.
- Deployment – Implementing the system within the university's server environment and configuring the application for its operational deployment.
- Maintenance & Updates – Continuously monitoring system performance, rectifying issues, and augmenting features in response to user feedback.

The project adheres to the Scrum framework, divided by 2-week sprints, thereby ensuring a systematic and manageable development process.

3) MODULES SPLIT-UP AND GANTT CHART

3.1 Modules Split-up

For the University Bursary Management System, the development process can be divided into the following modules:

Project Selection

- Define the project scope, focusing on digitizing the bursary management process.
- Identify pivotal objectives, including the enhancement of transparency, the minimization of manual tasks, and the facilitation of workflow monitoring.
- Determine the technology stack (Next.js employing TypeScript, Express.js, MySQL).

Project Planning

- Define project timeline, milestones, and deliverables.
- Assign responsibilities among team members.
- Use project management tools (e.g., Trello, Jira) to track progress and tasks.

Requirement Analysis

- Gather requirements from stakeholders, such as:
    - Bursary Department (Bursary MA, SAR)
    - Faculty Staff (AR, HOD, Department MA)
    - Students
- Identify system functionalities:
    - Bursary eligibility checking
    - Attendance workflow management
    - Issue/ticket handling
    - Academic calendar management
- Define non-functional requirements such as performance, security, and usability.

Learning Techniques

- Research frameworks and tools such as Express.js, Next.js, and MySQL.
- Study workflow-based system design and role-based access control.
- Explore best practices in secure authentication and system scalability.

Documentation

- Prepare initial documentation including:
    - System overview
    - Use case diagrams
    - DFDs (Context & Level 1)
    - ER diagrams
- Define functional and non-functional requirements clearly.

Design

- Design UI wireframes for:
    - Student dashboard
    - Staff dashboards
    - Issue management interface
- Design database schema for:
    - Students
    - Bursary records
    - Attendance sheets
    - Issues
    - Academic calendar
- Design system workflow (attendance approval flow across roles).
- Define API structure for frontend-backend communication.

Development

1. Frontend Development (Next.js with TypeScript)

- Develop user interfaces for:
    - Student portal
    - Staff dashboards
    - Tracking and issue systems
- Implement a responsive and user-friendly design.

2. Backend Development (Express.js)

- Implement business logic for:
    - Authentication and authorization
    - Workflow processing
    - Issue management
- Develop REST APIs and connect to MySQL databases.

3. Database Development (MySQL)

- Create and manage relational database tables.
- Ensure data integrity and efficient queries.

Testing and Correction

- Conduct:
    - Unit testing
    - Integration testing
    - User interface testing
- Identify and fix bugs.
- Optimize system performance.
- Perform **User Acceptance Testing (UAT)** with stakeholders such as bursary officers and faculty staff.

Implementation

- Deploy the system on the university server environment (Node.js compatible server).
- Ensure all system functionalities are working correctly after deployment.
- Configure system monitoring for performance and uptime.

Project Completion

- Finalize all documentation including:
    - User manual
    - Technical documentation
- Conduct final project evaluation and gather stakeholder feedback.
- Deliver the system and provide basic training for users (staff and students).

3.2 Gantt Chart

Table 1 Gantt Chart

Tasks	Time Duration (Weekly)														
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Project selection															
Project Planning															
Requirement Analysis															
Learning Techniques															
Documentation															
Design															
Development															
Test and correction															
Implementation															
Project Completion															

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5) COST ANALYSIS

Table 2 Cost analysis

Cost Item	Cost (for a month – LKR)
Hosting & Domain Costs	
Domain & Hosting	00.00 /= (University Subdomain)
Server Management Cost (Internet & Utilities)	
Internet Charges	2000/=
Electricity & Other Utilities	2500/=

6) APPROVAL LETTER FROM THE ORGANIZATION

To: Ms. Rameesha Madhurangi
Community Project Coordinator

Confirmation of Student Engagement in Bursary Management System Project

Dear Madam,

This is to formally confirm that the following students of the Department of Software Engineering, Faculty of Computing, Sabaragamuwa University of Sri Lanka, are currently engaged in the development of a *University Bursary Management System* as part of their academic and community project:

- 21CSE0157 – W.P.I.N. Kumarasinghe
- 21CSE0158 – L.B. Madhuranga
- 21CSE0171 – Ulindu Dakshitha
- 21CSE0184 – R.R. Wijekoon
- 21CSE0224 – Kavindu Dinal

This project is being carried out under my knowledge and coordination, in collaboration with the relevant university divisions, including the Bursary Branch.

It is important to note that bursary-related information involves highly sensitive student data. Therefore, granting full access and official authorization requires approval from higher authorities, including the Registrar, and this process is currently ongoing. As such, obtaining formal institutional approval may require additional time.

However, I would like to emphasize that the above-mentioned students are genuinely engaged in this project under the guidance of the Department of Software Engineering, and their work is being conducted strictly for academic purposes with due consideration to data confidentiality and ethical standards.

This letter is issued to support the submission of their project proposal for academic evaluation and community project requirements.

Thank you.

Yours sincerely,

Prof. B.T.G.S. Kumara
Head of Department
Department of Software Engineering
Faculty of Computing
Sabaragamuwa University of Sri Lanka

Figure 6: Approval letter from the organization